Econ 202B: Lecture 4

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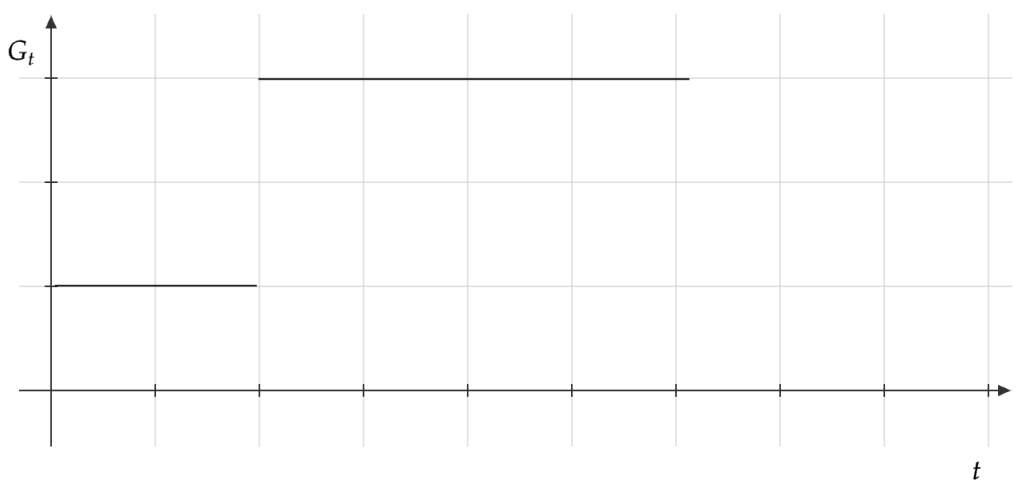
This lecture covers two counterfactuals – a permanent vs transitory shock to government expenditures. Using the RBC, we want to try to understand what happens to the aggregate economy given these shocks over time.

1 Permanent Shock

Consider a one-time, permanent increase in government expenditures.

Figure 1 Permanent Shock to G_t



1.1 Qualitative Assessment

The following table describes the direction of each variable given this one-time shock at different stages:

Table 1: Impact Evaluation

	Impact (relative to old s.s)	$t \to \infty$ (relative to impact)	New steady state (relative to old s.s)
λ	$+$	$-$	$+$
K	0	$+$	$+$
C	$-$	$+$	$-$
I	$+$	$-$	$+$
L	$+$	?	$+$
Y	$+$	?	$+$
w	$-$	$+$	0
R	$+$	$-$	0

1.2 Market Dynamics

We can deduce these results from studying the following three graphs:

Figure 2 Marginal Utility vs Capital

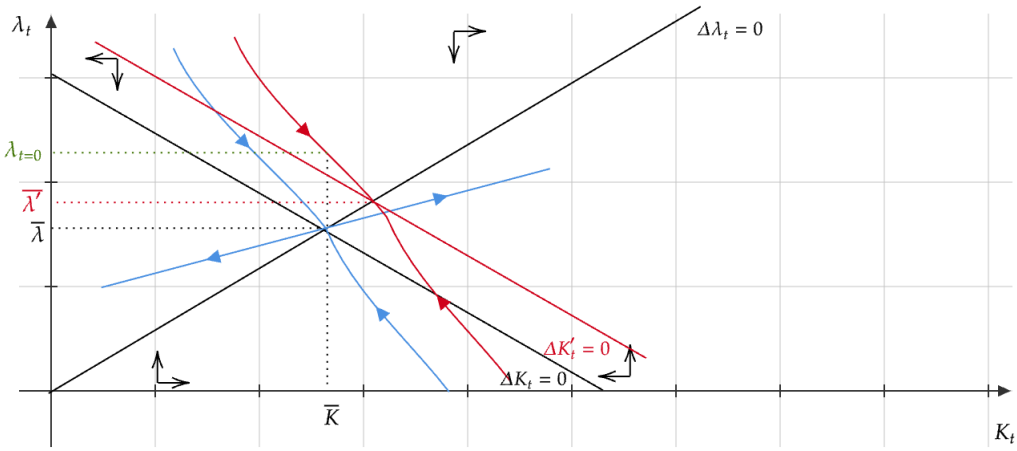


Figure 3 Labor market

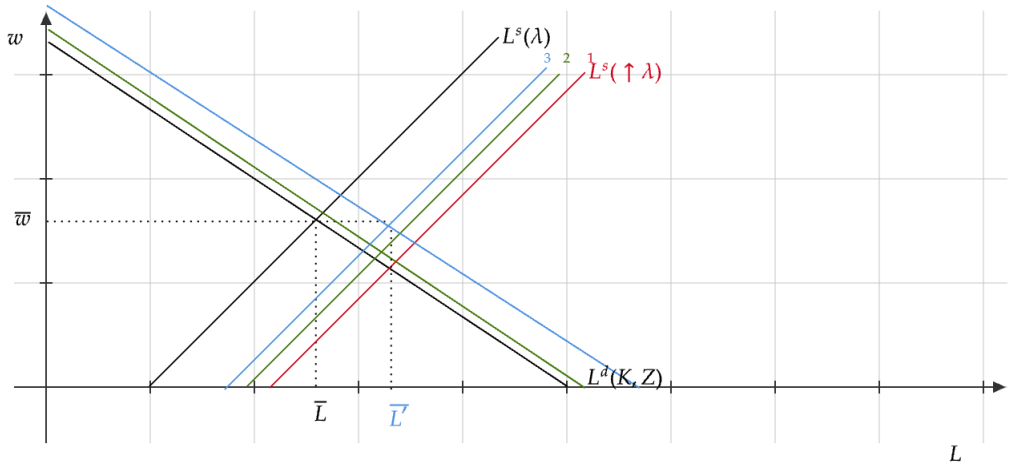
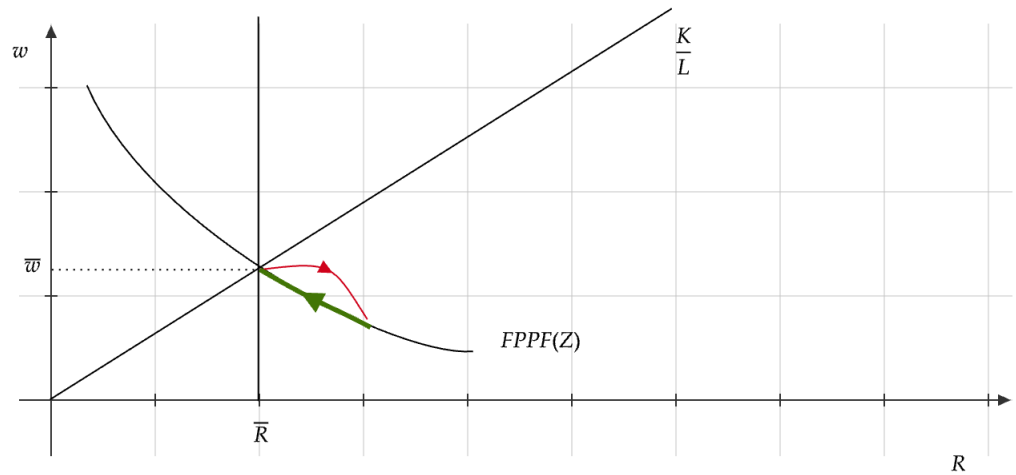
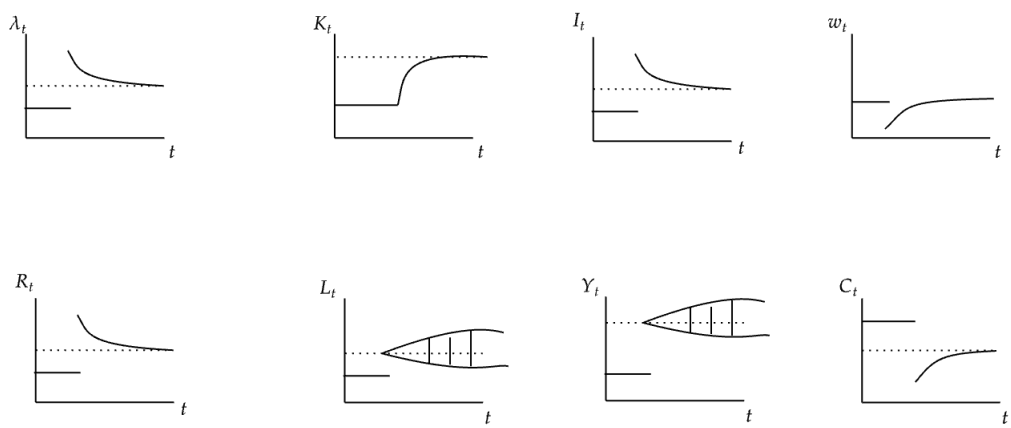


Figure 4 Factor Price Possibility Frontier (FPPF)



1.3 Impulse Response

Figure 5 Impulse Response



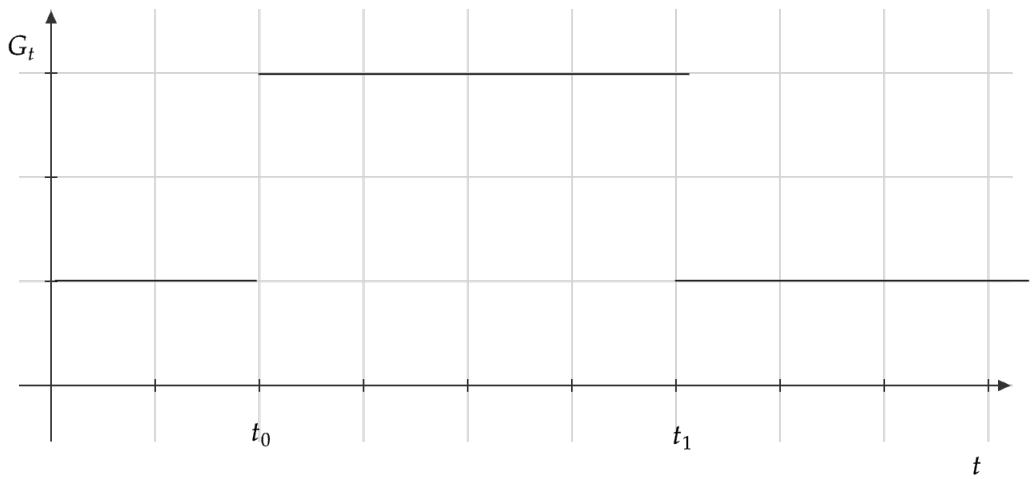
1.4 Summary

In the RBC model, government spending is always bad because consumption is always going down but we end up working more. This model also predicts a negative correlation with consumption and output which is at odds with the data. Therefore, we can only make two conclusions: either demand shocks are not the source of business cycles or the model is wrong. We will assume the former for now.

2 Transitory Shock

Now consider a transitory shock to government expenditures where agents have perfect information. The counterfactual shock will look like

Figure 6 Transitory Shock to G_t



2.1 Qualitative Assessment

The table from earlier will now have an additional column studying the dynamics at t_1 and the transition from t_1 to the new steady state. Agents have perfect information, so at t_0 they know that government expenditures will return to original steady state levels at t_1.

Table 2: Impact Evaluation

	Impact	$t_0 \to t_1$	t_1	$t_1 \to \infty$	Steady State
λ	$+$	$-$	0	$-$	0
K	0	$-$	0	$+$	0
C	$-$	$+$	0	$+$	0
I	$-$	$-$	$+$	$-$	0
L	$+$	$-$	0	?	0
Y	$+$	$-$	0	?	0
w	$-$	?	0	$+$	0
R	$+$	?	0	$-$	0

2.2 Market Dynamics

Consider the same market dynamics as before but with a transitory shock instead.

Figure 7 Marginal Utility vs Capital

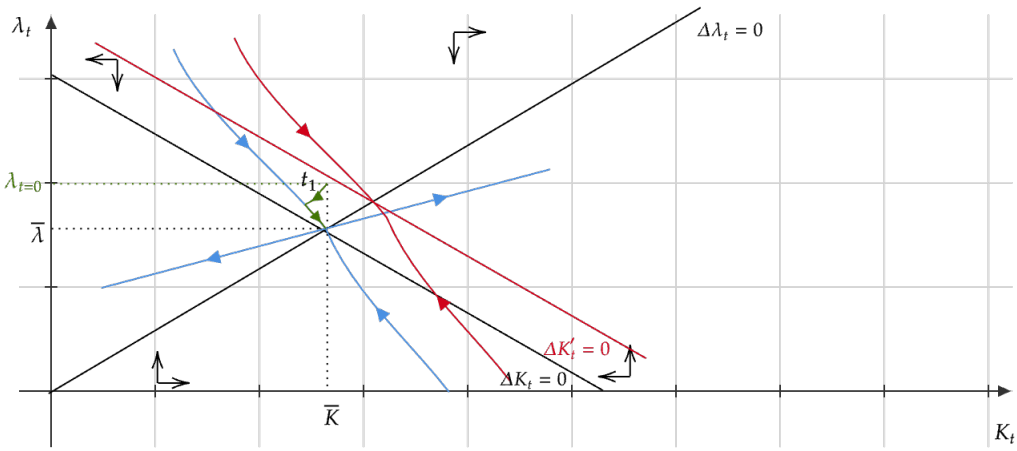


Figure 8 Labor market

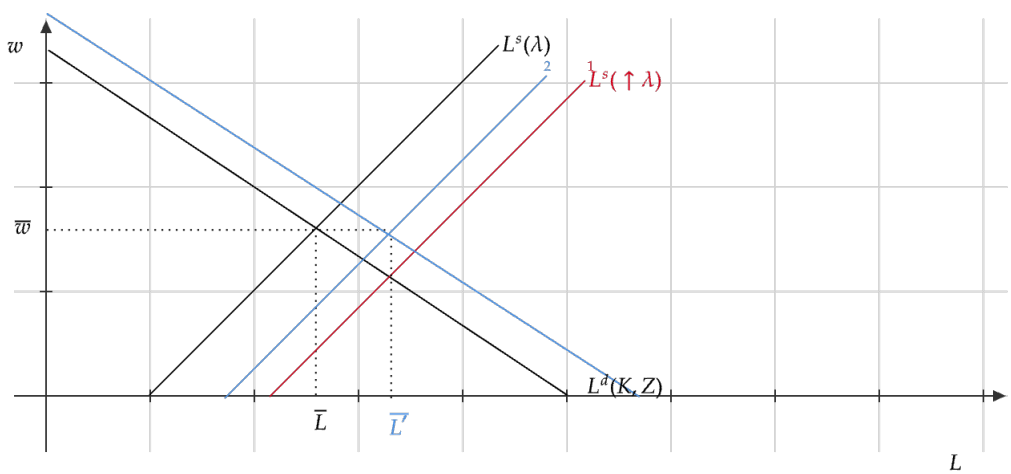
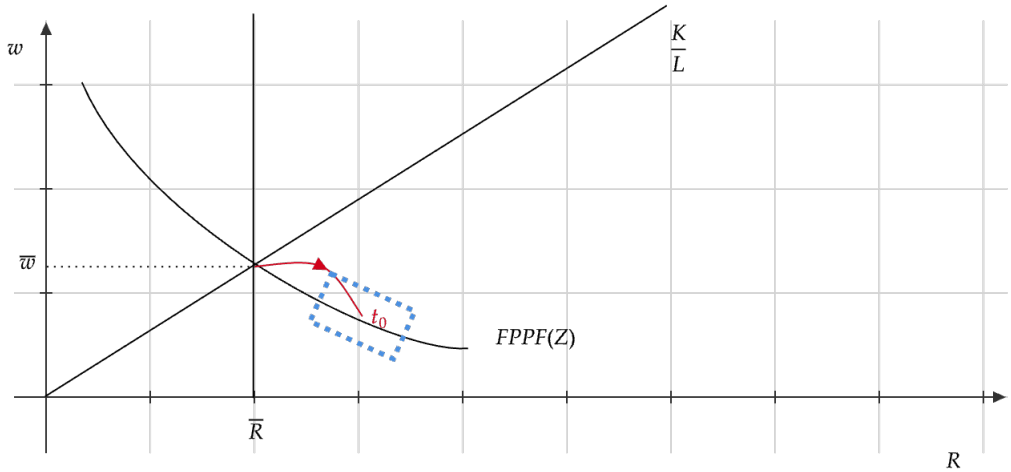
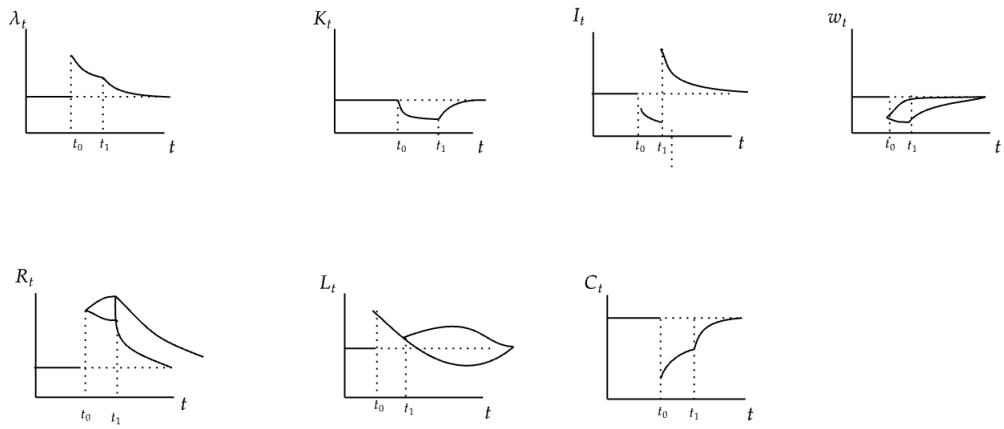


Figure 9 Factor Price Possibility Frontier (FPPF)



2.3 Impulse Response

Figure 10 Impulse Response



2.4 Summary

Again, we see a negative correlation between consumption and output. Even given a transitory shock, the model does not hold up with the data. Assuming the model is still fine, in the next lecture we will see how productivity shocks might change the system and whether this may be better explanation for the causes of business cycles.